



MATRICES MCQ — SUPER FAST TRICKS

Class 12 Applied Maths — Smart Solving Guide

Think of a matrix as a number table. MCQs mostly test patterns, not long solving. Goal = avoid calculation whenever possible.

⚡ TRICK 1: Order (Size) Check — Instant Elimination

Matrix order = Rows × Columns

If question asks $A + B$, addition is possible **ONLY** when orders are the same. If options include a different order → eliminate instantly. 50% of questions are solved this way.

⚡ TRICK 2: Matrix Multiplication Rule (Golden Rule)

Rule: $(m \times n) \times (n \times p) = (m \times p)$

Memory Trick: **Inside numbers must match.**

Example: $2 \times 3 \times 3 \times 4$ ✓ Possible | $2 \times 3 \times 2 \times 4$ ✗ Impossible.

⚡ TRICK 3: Identity Matrix Magic (BIGGEST TIME SAVER)

Identity Matrix (I) behaves like the number 1.

$$AI = IA = A$$

If an option shows AI , IA , or $A \rightarrow$ they are **ALL THE SAME**. No multiplication required 😎

⚡ **TRICK 4: Zero Matrix Shortcut**

Zero matrix = all elements 0. Multiplication with a zero matrix results in a zero matrix. Immediate answer!

⚡ **TRICK 5: Transpose Trick (Swap Swap!)**

Transpose = rows $\leftrightarrow$ columns swap. Formula: $(A^T)^T = A$

Formula: $(A+B)^T = A^T + B^T$

EXAM TRAP: $(AB)^T = B^T A^T$ (Order Reverses!)

Remember: **Transpose = Reverse Order**

TRICK 6: Symmetric vs Skew-Symmetric (1-Second Check)

Symmetric: $A^T = A$ (Mirror same)

Skew-Symmetric: $A^T = -A$

★ **SUPER SHORTCUT:** All diagonal elements = 0 in a skew-symmetric matrix. No need for a full transpose check!

TRICK 7: Equality of Matrices (FREE MARKS)

Two matrices are equal only if they have the **same order** and **corresponding elements are equal**. Compare just one unequal element from the options to eliminate wrong choices immediately.

TRICK 8: Scalar Multiplication

If matrix multiplied by number k , every element $\times k$. **Hack:** Check only one element instead of the full matrix to save time.

TRICK 9: Determinant-Type Pattern Guess

If rows or columns are proportional (e.g., Row 2 = 2 $\times$ Row 1), the **determinant = 0**. No calculation needed.

TRICK 10: Option Substitution (TOPPER SECRET)

Instead of solving: Put the option value into the equation. Check which satisfies the condition. MCQs become 5-second questions.

5-SECOND EXAM STRATEGY

1. Check order
2. Look for identity/zero matrix
3. Check transpose rules
4. Eliminate impossible options
5. Substitute options if needed

Solve mentally before writing anything!

MEMORY CODE (Revise Before Exam)

O I Z T S E S D O

Order | Identity | Zero | Transpose | Symmetric | Equality | Scalar | Determinant
Pattern | Option Check

By Thunderstudy